

E74984_53

The equation $-x^4 + x^2 + xy + y^3 + 3 = 0$ has no integer solutions

Size $h = 35$, length $l = 12.58$

Theorem 1. *The equation*

$$-x^4 + x^2 + xy + y^3 + 3 = 0 \tag{1}$$

has no integer solutions.

The proof uses two polynomial identities of the form $H_k T_k = U_k^2 - 3V_k^2$ on the curve (1), and the fact that an odd prime p with $(3/p) = -1$ divides $U_k^2 - 3V_k^2$ only to an even power. Apart from polynomial identities, which can be checked by expanding both sides, and finite checks of congruences, the proof uses only basic properties of the Jacobi symbol and the law of quadratic reciprocity.

1 Jacobi symbols and the norm form $a^2 - 3b^2$

For a prime p and a non-zero integer N , we write $v_p(N)$ for the exponent of p in the prime factorization of N . For an integer a and an odd prime p , the Legendre symbol (a/p) is 0 if $p \mid a$, 1 if a is a non-zero square modulo p , and -1 otherwise. For an odd positive integer n , the Jacobi symbol is $(a/n) = \prod_p (a/p)^{v_p(n)}$, the product over the prime factors p of n . It is multiplicative in a and in n , it depends only on a modulo n , and it is non-zero exactly when a and n are coprime. We use the law of quadratic reciprocity for Jacobi symbols together with its supplements [1, Chapter 5, §2]: for odd positive integers m and n ,

$$\begin{aligned} \left(\frac{-1}{n}\right) &= (-1)^{(n-1)/2}, & \left(\frac{2}{n}\right) &= (-1)^{(n^2-1)/8}, \\ \left(\frac{m}{n}\right) \left(\frac{n}{m}\right) &= (-1)^{(m-1)(n-1)/4} \quad \text{if } \gcd(m, n) = 1. \end{aligned}$$

Lemma 2. *Let p be an odd prime with $(3/p) = -1$, and let $a, b \in \mathbb{Z}$. If $p \mid a^2 - 3b^2$, then $p \mid a$ and $p \mid b$.*

Proof. Suppose that $p \nmid b$. Choose $c \in \mathbb{Z}$ with $bc \equiv 1 \pmod{p}$. Then $(ac)^2 \equiv 3(bc)^2 \equiv 3 \pmod{p}$, so 3 is a square modulo p . As $(3/p) \neq 0$, it is a non-zero square, so $(3/p) = 1$, a contradiction. Hence $p \mid b$, and then $p \mid a^2$, so $p \mid a$. $\square$

Lemma 3. *Let $a, b \in \mathbb{Z}$ be not both zero. Then $a^2 - 3b^2 \neq 0$, and $v_p(a^2 - 3b^2)$ is even for every odd prime p with $(3/p) = -1$.*

Proof. Since 3 is not the square of an integer, $a^2 - 3b^2 \neq 0$. Let p be an odd prime with $(3/p) = -1$, let p^k be the largest power of p dividing both a and b , and write $a = p^k a'$ and $b = p^k b'$. Then p does not divide both a' and b' , so $p \nmid a'^2 - 3b'^2$ by Lemma 2. Since $a^2 - 3b^2 = p^{2k}(a'^2 - 3b'^2)$, we get $v_p(a^2 - 3b^2) = 2k$. $\square$

Lemma 4. *Let n and n' be odd positive integers with $n' \equiv n \pmod{12}$, or with $n' \equiv -n \pmod{12}$. Then $(3/n') = (3/n)$.*

Proof. Write $3 = \varepsilon 2^e q$ with $\varepsilon = 1$, $e = 0$ and $q = 3$. By multiplicativity and the reciprocity laws,

$$\left(\frac{3}{n}\right) = \left(\frac{\varepsilon}{n}\right) \left(\frac{2}{n}\right)^e (-1)^{(n-1)(q-1)/4} \left(\frac{n}{q}\right).$$

Here (ε/n) and $(-1)^{(n-1)(q-1)/4}$ depend only on n modulo 4, $(2/n)$ depends only on n modulo 8, and (n/q) depends only on n modulo q . Since 12 is divisible by $4q$, the first claim follows. If $n' \equiv -n \pmod{12}$, then, as $\varepsilon = 1$, the factor (ε/n) does not change, $(2/n)$ does not change because $n'^2 \equiv n^2 \pmod{16}$, and each of $(-1)^{(n-1)(q-1)/4}$ and (n/q) is multiplied by $(-1)^{(q-1)/2}$, because $(n' - 1)/2 \equiv (n - 1)/2 + 1 \pmod{2}$ and $(n'/q) = (-1/q)(n/q) = (-1)^{(q-1)/2}(n/q)$. This proves the second claim. $\square$

Lemma 5. *Let n be a non-zero integer, and let s , u and m be positive integers such that every prime factor of s divides 6, both $2s$ and $12s$ divide m , u is odd, $(3/u) = -1$, and $n \equiv su \pmod{m}$. Then there is an odd prime p with $(3/p) = -1$ such that $v_p(n)$ is odd.*

Proof. Since $s \mid m$, we have $s \mid n$, and $n' = n/s$ satisfies $n' \equiv u \pmod{m/s}$. Since m/s is even and divisible by 12, n' is odd and $n' \equiv u \pmod{12}$. Since $|n'| \equiv \pm u \pmod{12}$, Lemma 4 gives $(3/|n'|) = (3/u) = -1$. By multiplicativity, the product of $(3/p)^{v_p(n')}$ over the prime factors p of $|n'|$ is -1 , so some prime factor p of n' satisfies $(3/p) = -1$ and $v_p(n')$ is odd. Then p is odd, and $p \nmid 6$, so $p \nmid s$ and $v_p(n) = v_p(n')$ is odd. $\square$

2 The auxiliary polynomials

Put $F(x, y) = -x^4 + x^2 + xy + y^3 + 3$, so that (1) is the equation $F(x, y) = 0$. For the first identity, we define

$$\begin{aligned} H_1 &= -2x + y - 2, \\ T_1 &= 4x^2 + 2xy + 9x + y^2 + 2y + 4, \\ U_1 &= x^2 - 4x - 8, \\ V_1 &= 3x + 5, \\ G_1 &= 1, \\ A_1 &= 9, \\ B_1 &= -3x + 17, \end{aligned}$$

for the second identity, we define

$$\begin{aligned} H_2 &= 4x + y + 4, \\ T_2 &= 16x^2 - 4xy + 33x + y^2 - 4y + 16, \\ U_2 &= x^2 + 32x + 19, \\ V_2 &= 17x + 10, \\ G_2 &= 1, \\ A_2 &= 289, \\ B_2 &= -17x - 534. \end{aligned}$$

Expanding both sides, one checks the identities

$$H_1 T_1 - U_1^2 + 3V_1^2 = G_1 F, \tag{2}$$

$$H_2 T_2 - U_2^2 + 3V_2^2 = G_2 F, \tag{3}$$

$$A_1 U_1 + B_1 V_1 = 13, \tag{4}$$

$$A_2 U_2 + B_2 V_2 = 151. \tag{5}$$

3 Proof of Theorem 1

Suppose that $(x, y) \in \mathbb{Z}^2$ is a solution of (1). Then $F(x, y) = 0$, and (2) and (3) give

$$H_k T_k = U_k^2 - 3V_k^2 \quad (k = 1, 2), \quad (6)$$

where the polynomials now denote their values at (x, y) .

Step 1: congruences. Checking the 36^2 residue pairs modulo 36, we find that (1) has exactly 24 solutions modulo 36, and that at each of them $H_1 \bmod 36$ lies in the set $\mathcal{R}_1$ listed in Table 1 or $H_2 \bmod 36$ lies in the set $\mathcal{R}_2$ listed in Table 1. For $k = 1$, each $r \in \mathcal{R}_1$ is listed as $r = su$ with s and u positive and u odd, or as r if $s = 1$, and in each case $(3/u) = -1$; here for $s = 1$, $2s = 2$ and $12s = 12$ divide 36, and for $s = 3$, every prime factor of s divides 6, and $2s = 6$ and $12s = 36$ divide 36. For $k = 2$, each $r \in \mathcal{R}_2$ is listed as $r = su$ with s and u positive and u odd, or as r if $s = 1$, and in each case $(3/u) = -1$; here for $s = 1$, $2s = 2$ and $12s = 12$ divide 36, and for $s = 3$, every prime factor of s divides 6, and $2s = 6$ and $12s = 36$ divide 36.

Step 2: signs. *The value of H_1 .* By (4), U_1 and V_1 are not both zero, so $U_1^2 - 3V_1^2 \neq 0$ by Lemma 3, and (6) gives $H_1 \neq 0$.

The value of H_2 . By (5), U_2 and V_2 are not both zero, so $U_2^2 - 3V_2^2 \neq 0$ by Lemma 3, and (6) gives $H_2 \neq 0$.

Step 3: primes p with $(3/p) = -1$. Let $k \in \{1, 2\}$, and let p be an odd prime with $(3/p) = -1$. We show that $v_p(H_k)$ is even. By (4)–(5), U_k and V_k are not both zero, so (6) and Lemma 3 show that $v_p(H_k T_k) = v_p(U_k^2 - 3V_k^2)$ is even. Suppose that p divides both H_k and T_k . Then $p \mid U_k^2 - 3V_k^2$ by (6), so $p \mid U_k$ and $p \mid V_k$ by Lemma 2, and $p \mid R_k$, where R_k is the right-hand side of the identity for $A_k U_k + B_k V_k$. For $k = 1$, we have $R_1 = 13$. But p is odd, and $p \neq 13$ because $(3/13) = 1$, a contradiction. For $k = 2$, we have $R_2 = 151$. As p is odd, we get $p = 151$. Checking the 151^2 residue pairs modulo 151, we find that F , H_2 and T_2 have no common zero modulo 151, a contradiction. Hence p does not divide both H_k and T_k . If $p \nmid T_k$, then $v_p(H_k) = v_p(H_k T_k)$ is even, and if $p \nmid H_k$, then $v_p(H_k) = 0$.

Step 4: contradiction. By Step 1, there is k such that $H_k \equiv su \pmod{36}$ with s and u as in Lemma 5, and $H_k \neq 0$ by Step 2. Hence Lemma 5 applies to $n = H_k$, so there is an odd prime p with $(3/p) = -1$ such that $v_p(H_k)$ is odd. This contradicts Step 3, and the theorem is proved. $\square$

References

- [1] K. Ireland and M. Rosen, *A Classical Introduction to Modern Number Theory*, 2nd ed., Graduate Texts in Mathematics 84, Springer, New York, 1990.

Table 1: The sets $\mathcal{R}_k$ of residues modulo 36 in Step 1.

$\mathcal{R}_1$, the residues of H_1						
5	7	17	19	$21 = 3 \cdot 7$	29	31
$\mathcal{R}_2$, the residues of H_2						
5	7	17	19	$21 = 3 \cdot 7$	29	31